

Same-well imaging and transcriptomics in a 50,000-well one-bead-one-compound microchip recover compound-resolved phenotypes

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Abstract

Phenotypic screens usually buy scale by giving up biological depth, or buy depth by giving up compound throughput. High-content imaging can profile very large libraries, but the readout is morphological. Single-cell transcriptomics gives a deeper view of cell state, but at a higher cost per treatment. ActiveSeq was built to test a narrower and more practical question: can the same Z-Screen micro-well provide a useful paired image and RNA readout from the same 5 to 10 cell colony?

We analyzed 20,222 matched wells from two HEK293 pilot experiments, ACT010 and ACT011. The dataset contains 11,511 wells from a 35-compound control panel and 8,711 ZEL026 dummy-bead wells used as a negative-control QC stress test. The image representation used for the main analyses is an internally calibrated 32-dimensional latent learned from a frozen per-well image embedding on this same control-panel domain. It should therefore be read as a calibrated assay representation, not as evidence that the uncalibrated vision embedding generalizes to new chemistry or cell types.

At the compound level, both readouts were reproducible. Split-half top-1 retrieval reached 98.8% for RNA and 94.2% for the calibrated image latent in ACT010, and 80.1% / 86.8% in ACT011, against 2.9% chance. ACT010 centroids retrieved the same ACT011 compounds at 71.4% top-1 for RNA and 85.7% top-1 for image. Image and RNA also preserved related, non-identical compound maps, with cross-modal distance correlations of 0.718 in ACT010 and 0.607 in ACT011.

The strongest operational result was the held-out classification gain from combining image and RNA. A linear discriminant trained on merged image-plus-RNA features reached 55.5% balanced accuracy, compared with 41.4% for RNA alone and 29.2% for image alone; the same ordering held within both runs. A same-compound shuffled-fusion control matched true same-well fusion on this task, so the supervised gain should be read as multimodal compound-level additivity. Analyses that ask the stricter same-well question found positive but modest within-compound structure and should be treated as exploratory support.

The RNA branch remained interpretable at low pass depth. Centered pseudobulk gene signatures retrieved the correct compound across runs at 48.6% top-1 and 68.6% top-3, with compound-specific gene programs suitable for triage and follow-up hypothesis generation. Relative to scGeneScope, the

closest public treatment-matched Cell Painting and scRNA-seq reference, ActiveSeq is much smaller and uses a simpler imaging modality. After normalizing for label-space difficulty, its multimodal classification effect falls in the same range, which provides magnitude context rather than a direct benchmark victory.

Significance

Treatment-matched Cell Painting and Perturb-seq experiments describe separate cell populations exposed to the same perturbation. ActiveSeq instead reads image and RNA from the same multicellular colony in the same micro-well, on a chip designed for 50,000 wells in parallel. In this pilot, both readouts recover compound identity, their maps are aligned but not redundant, and their merged representation improves held-out classification across the 35-compound panel. For early discovery, the useful claim is not that same-well pairing is already fully proven in every sense; it is that a paired well can carry enough orthogonal image and RNA information to make triage more informative than either readout alone.

Introduction

Early-stage discovery needs enough throughput to explore chemical space and enough biological resolution to decide whether a hit is worth pursuing. Existing platforms usually emphasize one side. Cell Painting and imaging-first ML systems can run at very large scale, but they infer biology from morphology [1]. Perturb-seq and related transcriptomic resources provide deeper cell-state measurements, but at lower compound concurrency and higher cost per profile [3]. Public multimodal datasets bridge the two by matching treatments across replicate cell populations; scGeneScope is the strongest current public example, pairing Cell Painting and scRNA-seq across treatments in U2-OS cells [4].

Z-Screen asks whether the pairing can be made physically tighter. In ActiveSeq, a one-bead-one-compound micro-well is imaged and then sequenced, so each well contributes one image vector and one low-pass RNA vector from the same 5 to 10 cell colony. That design does not remove all biological heterogeneity, and this pilot does not yet prove that exact same-well pairing is superior to every treatment-level fusion strategy. It does let us ask whether a single well carries two reproducible, aligned, and operationally additive descriptions of compound response.

The Results follow the decision sequence a skeptical screen designer would use: 1. Did the matched wells and negative controls behave well enough to analyze, and what exactly is the calibrated image representation? 2. Does each modality recover compound identity reproducibly? 3. Are the image and RNA compound maps aligned without being redundant? 4. Does combining the two readouts improve a held-out operational task? 5. Is there evidence for additional same-well structure beyond compound labels, and how large is it? 6. Does low-pass RNA remain interpretable enough for biological triage? 7. How large are these effects relative to scGeneScope, without treating that comparison as a direct win/loss benchmark?

Results

Matched wells pass QC, and image calibration defines the representation tested here

Before interpreting multimodal signal, we first asked whether the matched wells and assay controls behaved as expected. The dataset contained 20,222 matched wells from ACT010 and ACT011: 11,511 wells assigned to the 35-compound control panel and 8,711 ZEL026 dummy-bead wells. ZEL026 dummy wells contain no compound and were retained as a negative-control assay-chemistry stress test, not as a biological discovery condition. They showed the expected collapse in RNA yield: median UMIs dropped from 5,167 to 342 in ACT010 and from 4,750 to 293 in ACT011; detected genes dropped from 2,826 to 301 and from 2,463 to 248. This failure mode is visible in the readout itself and is reported as Supplementary Figure 1.

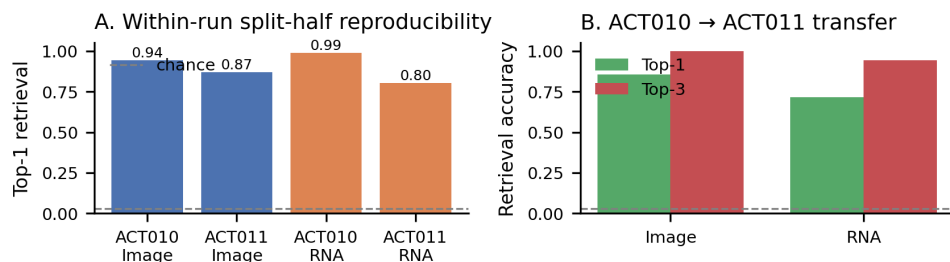
The next question was whether the image branch had a representation suitable for compound-level analysis. The original per-well image embedding came from a frozen pretrained vision model. In raw form, such an embedding can be dominated by experiment, position, or microscopy effects rather than compound response. We therefore used a 32-dimensional domain-adversarial latent calibrated on the matched control-panel domain. The calibration objective rewards compound discrimination while penalizing experiment-of-origin discrimination. The calibrated image latent is consequently an internal assay representation: it can reveal compound-resolved image signal in this pilot, but it is not yet evidence of off-panel image generalization.

With QC and representation scope defined, the first biological question was whether each modality reproducibly encoded the 35-compound panel.

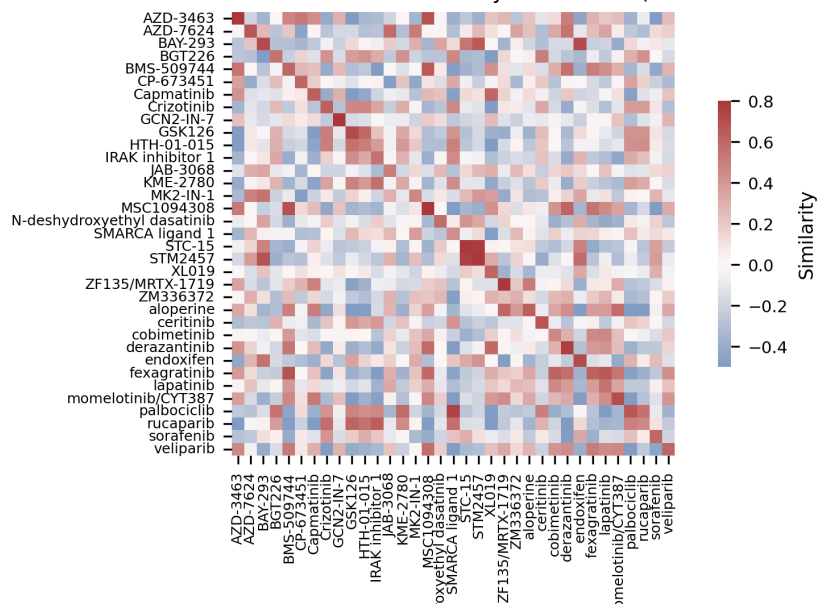
Each modality reproducibly encodes compound identity

Wells for each control compound were split into two pseudo-replicates, and one half was queried for retrieval of the other among the 35 compounds. RNA latent space was highly stable: split-half top-1 retrieval was 98.8% in ACT010 and 80.1% in ACT011. The calibrated image latent was similar at 94.2% in ACT010 and 86.8% in ACT011. Top-3 retrieval reached 100% for RNA in ACT010 and 99.6% for the calibrated image latent.

Cross-run transfer is the stricter test of a control-panel measurement. Compound centroids learned in ACT010 retrieved the same compounds in ACT011 at 71.4% top-1 (94.3% top-3) in RNA and 85.7% top-1 (100% top-3) in the calibrated image latent, against a 2.9% top-1 chance level. These results show compound-level reproducibility and transferability. They do not yet show that exact same-well pairing carries extra information, because split-half and centroid retrieval are driven by compound labels aggregated across many wells. That stricter pairing question is addressed later.



C. RNA centroid similarity across runs (ACT010 rows × ACT011 cols)



D. Image centroid similarity across runs

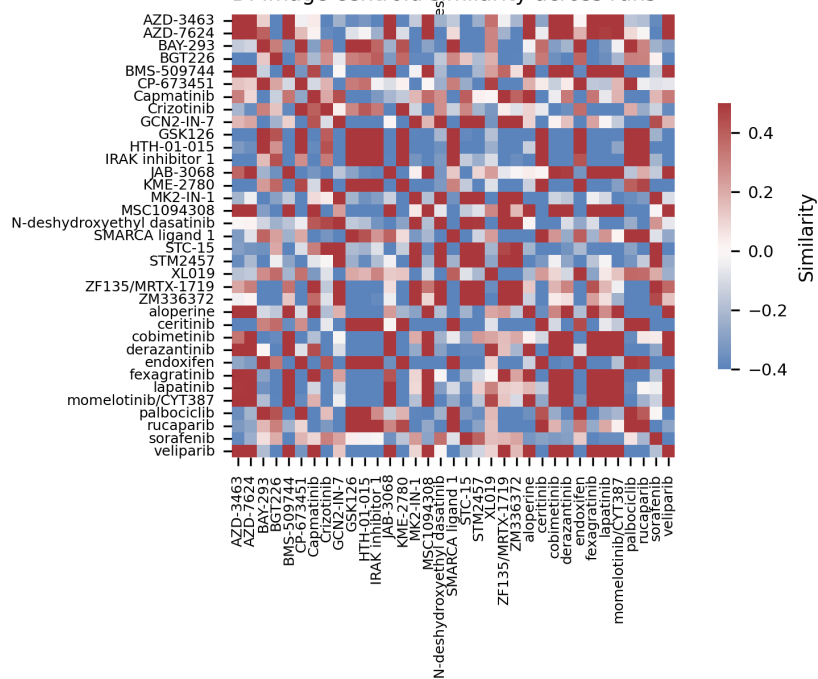


Figure 1. Compound-level reproducibility and cross-run centroid transfer in the 35-compound control panel. Both modalities are reproducible within run, and both transfer well above chance from ACT010 to ACT011. These metrics establish compound-level signal, not the incremental value of exact same-well pairing.

Image and RNA share a compound map but are not redundant

If both modalities recover compound identity, the next question is whether they organize compounds in the same way. Exact label recovery alone could hide two unrelated representations that happen to separate the 35 classes. We therefore compared pairwise compound distances in image and RNA space within each run.

The two maps were aligned. Spearman correlations between pairwise compound-distance matrices were 0.718 in ACT010 and 0.607 in ACT011. The largest perturbations also recurred across experiments: cross-run effect-magnitude rankings correlated at Spearman 0.799 in image space and 0.662 in RNA space. Several compounds, including GCN2-IN-7, KME-2780, BGT226, ZF135 / MRTX-1719, and CP-673451, appeared among the stronger responses in both modalities in at least one run.

The caveat is important for triage: a shared map is not an identical map. The image latent and RNA latent agree enough that image-derived neighborhoods are biologically meaningful, but they preserve different aspects of response. That nonredundancy motivates a direct test of whether combining them improves an operational task.

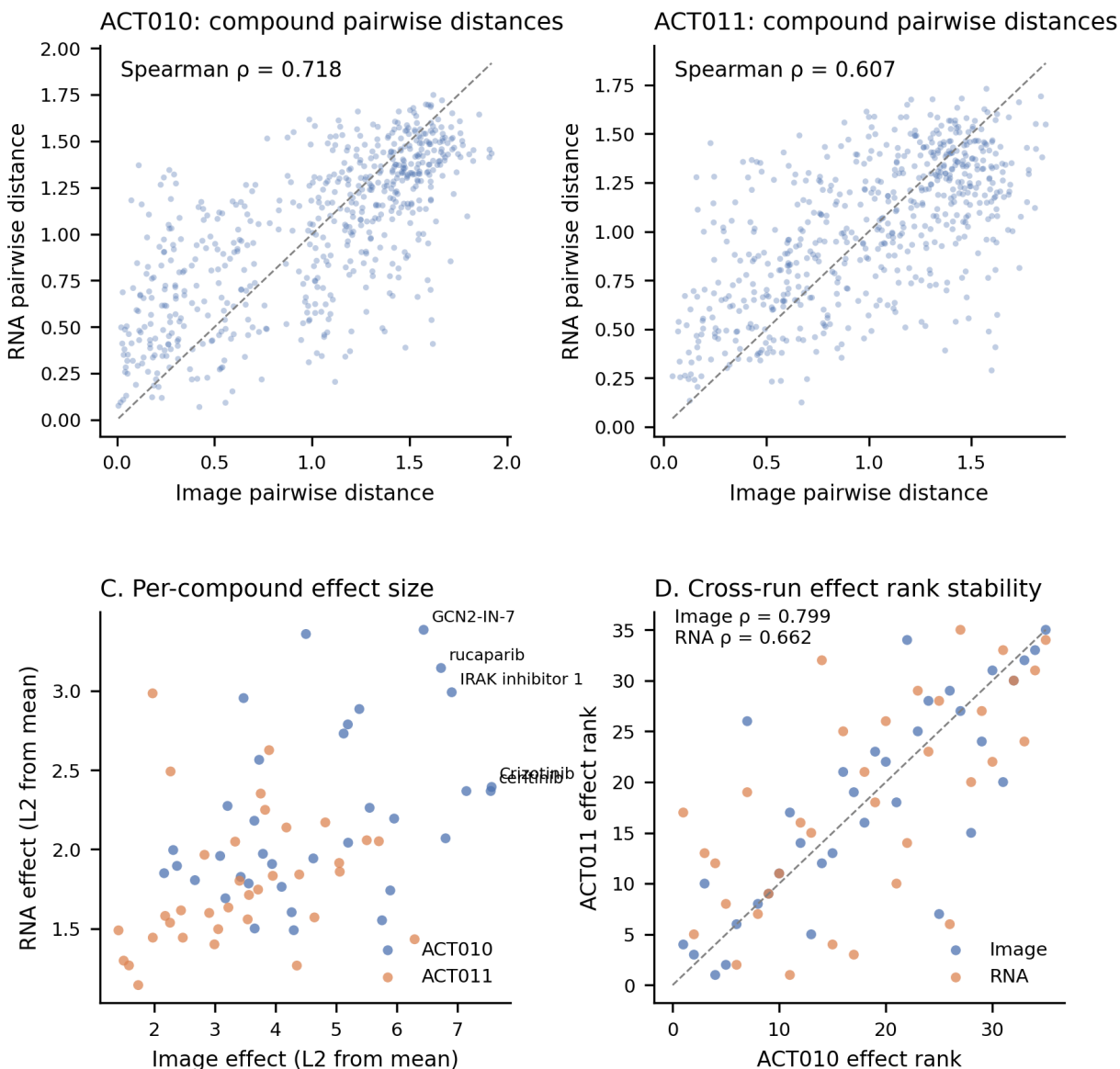


Figure 2. Image and RNA preserve a shared but non-identical compound geometry within run, and effect-size rankings are stable across runs. Pairwise compound-distance correlations are 0.718 in ACT010 and 0.607 in ACT011; cross-run per-compound effect magnitudes correlate at Spearman 0.799 for image and 0.662 for RNA.

Combining same-well image and RNA improves held-out compound classification

The practical question is whether the two same-well readouts help a decision model more than either readout alone. Each matched well was reduced to one calibrated image vector and one RNA vector. A linear discriminant model was trained on image alone, RNA alone, or the raw concatenation of image plus RNA, then evaluated on held-out wells.

Merged image-plus-RNA features outperformed RNA alone in every cut: 55.5% balanced accuracy

versus 41.4% pooled, 60.8% versus 47.3% within ACT010, and 45.1% versus 30.5% within ACT011. Image alone reached 29.2% balanced accuracy pooled. Overall accuracy followed the same pattern, rising from 42.5% to 54.9% pooled, from 48.3% to 60.2% in ACT010, and from 31.6% to 45.5% in ACT011.

The gain was broad rather than concentrated in a few easy classes. Merged features improved per-compound recall for 35 of 35 compounds in the pooled analysis, 33 of 35 in ACT010, and 30 of 35 in ACT011. ACT010 confusion matrices show the same pattern: RNA performs well on its own, and adding the calibrated image latent moves probability mass onto correct diagonal entries and resolves several near-confusions.

This is the strongest evidence in the paper that the paired readout is useful for compound classification. It should still be interpreted precisely. A same-compound unpaired-fusion control, in which image and RNA vectors were shuffled among wells of the same compound and run before fitting the same classifier, matched the true same-well merged result almost exactly. Across 100 shuffles, pooled balanced accuracy averaged 55.4% (95% interval 54.7% to 56.1%), compared with 55.5% for true same-well fusion. ACT010 and ACT011 showed the same pattern. The supervised gain therefore reflects multimodal compound-level additivity in the ActiveSeq readout, not an isolated contribution of exact well-to-well pairing on this label task.

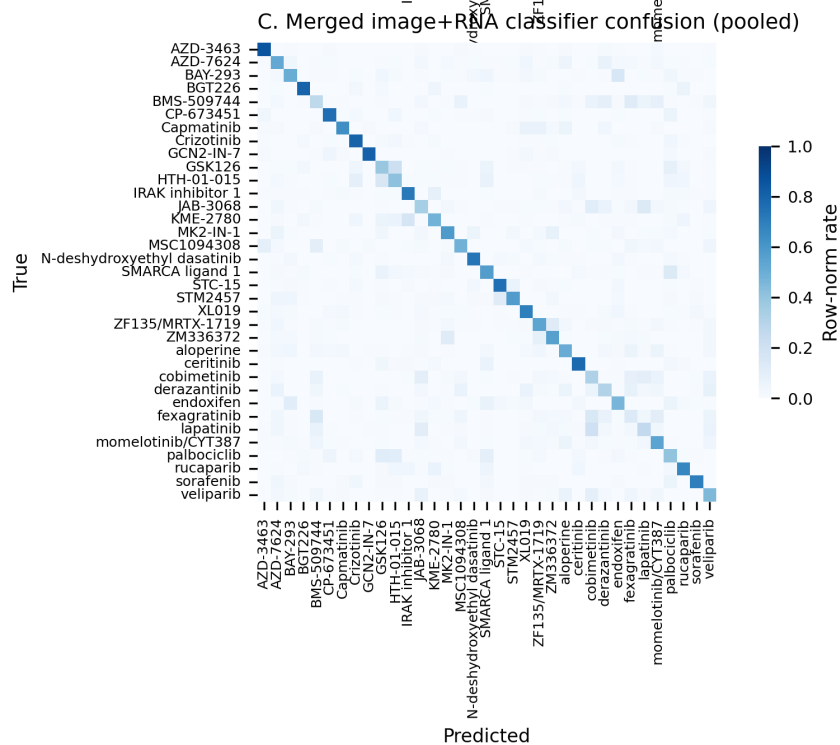
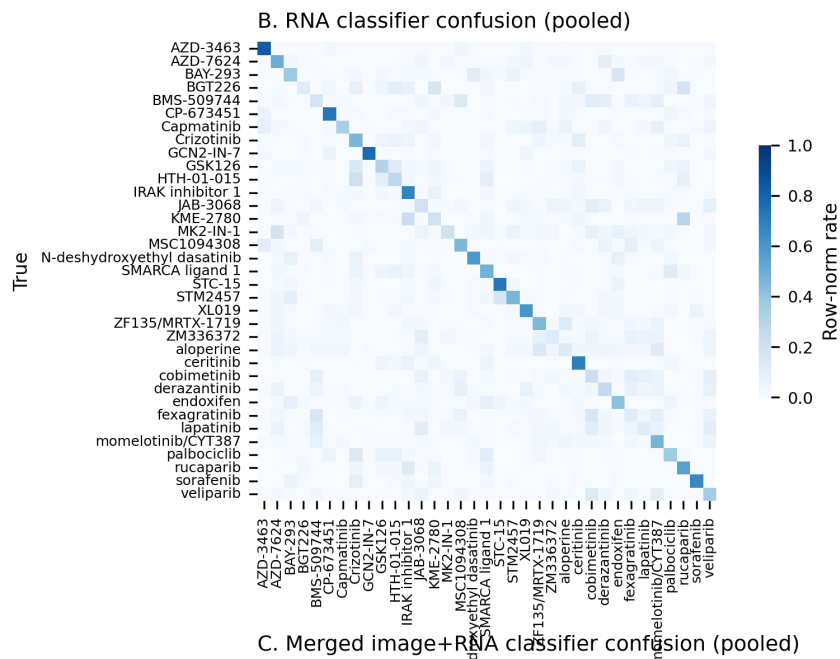
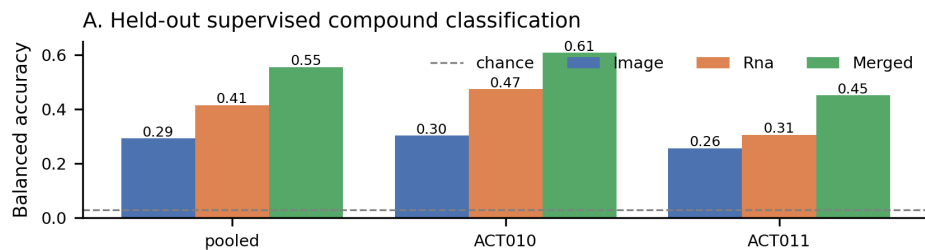


Figure 3. Same-well multimodal classification on held-out wells. Merged image-plus-RNA features outperform RNA alone in pooled and per-run analyses, and per-compound confusion matrices show the gain is from reduced class confusion across most compounds rather than from a few easy classes.

Within-compound same-well structure is positive but modest

The remaining question is the stricter same-well one: after compound identity is held constant, do image and RNA preserve matching well-to-well structure? This matters because exact same-well pairing would be most valuable if it captured response-state differences among wells exposed to the same compound, not only the average compound effect.

In the harder held-out centroid-retrieval task, the calibrated image latent reached 25.1% / 21.7% top-1 across the two runs, RNA reached 44.8% / 24.3%, and a naive image-plus-RNA concatenation reached 49.1% / 33.7%, exceeding RNA alone in both runs. A lightweight CLIP-style same-well contrastive objective on top of the calibrated latent did not improve retrieval further, indicating that calibration had already exposed most of the recoverable compound-identity information.

The exact same-well signal was above random but small. Top-10 exact image-to-RNA retrieval after contrastive training was 2.1% in ACT010 and 2.5% in ACT011, compared with random expectations of 0.4% and 1.0%. Within each compound, image and RNA distance structure was positive in 34 of 35 compounds in both runs, but the median cross-modal distance correlations were only 0.087 in ACT010 and 0.055 in ACT011. Mean kNN overlap gains were 0.006 and 0.020. These effects are compatible with real within-compound response-state structure, but they are not large enough to carry the main claim. The shuffled-fusion control above reinforces that boundary: exact same-well pairing is visible in exploratory within-compound summaries, but the current supervised classification gain is driven by compound-level multimodal information.

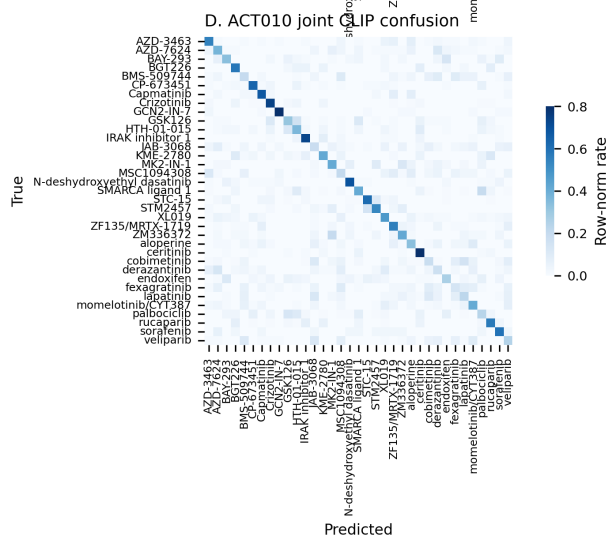
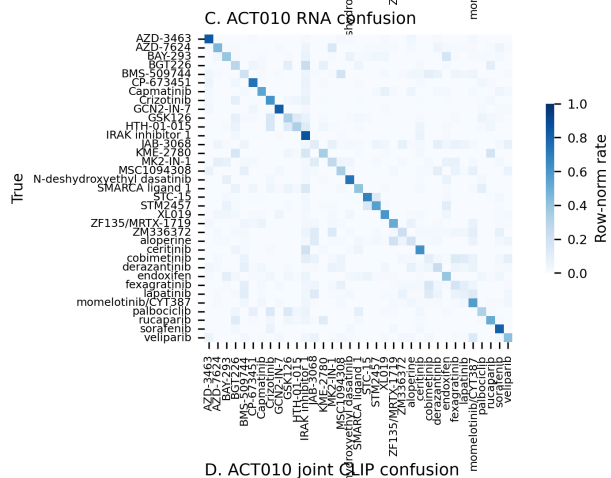
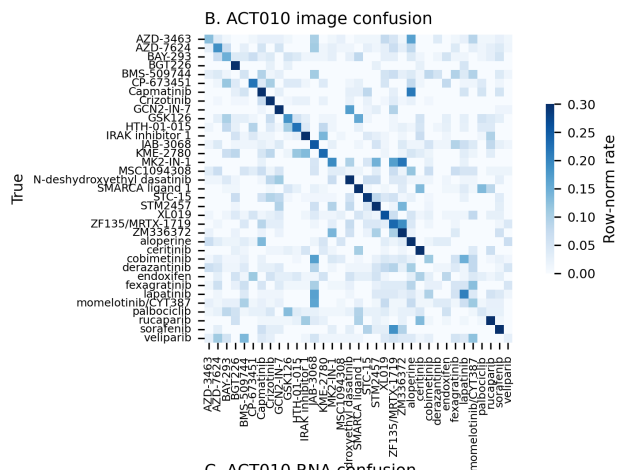
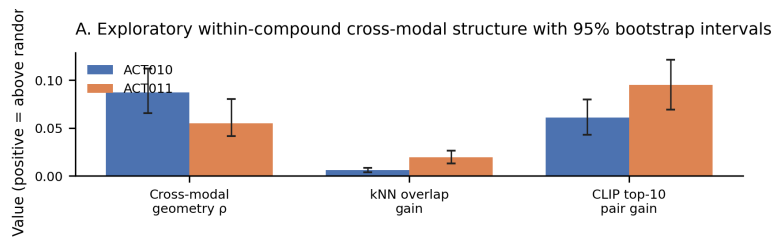


Figure 4. Exploratory same-well structure beyond supervised classification. Within-compound cross-modal agreement is usually above random but modest in magnitude. These data support the possibility of same-well response-state information, while leaving the incremental value of exact pairing to a dedicated control analysis.

Low-pass RNA remains interpretable enough for triage

The classification results say that RNA helps prediction. Drug hunters also need to know whether the RNA readout remains interpretable enough to guide follow-up. We therefore aggregated raw counts from matched control wells into per-compound pseudobulk signatures, converted them to log-CPM, and centered each run by subtracting its experiment-wide mean. This deliberately treats the compound-by-run pseudobulk as the robust unit, rather than overfitting per-well differential expression at low RNA depth.

Centered signatures retrieved the correct compound across runs at 48.6% top-1 and 68.6% top-3. Representative compounds carried coherent gene programs, including palbociclib (ARC, RASD1, CDKN1C, EGR1), BAY-293 (MT1X, MT1F, MT2A, SLC30A1), rucaparib (ITPRIP, SLC37A4, NACC2, COL4A1), crizotinib (SORCS2, ZNF34, AC253572.2), and GSK126 (DUSP10, SPATA6L, CCN2). These examples should be read as interpretable rank signatures for triage, not as validated mechanisms of action in HEK293. The useful result is that low-pass ActiveSeq RNA preserves enough compound-specific gene structure to cluster hits, flag stress or metal-response programs, and nominate follow-up biology.

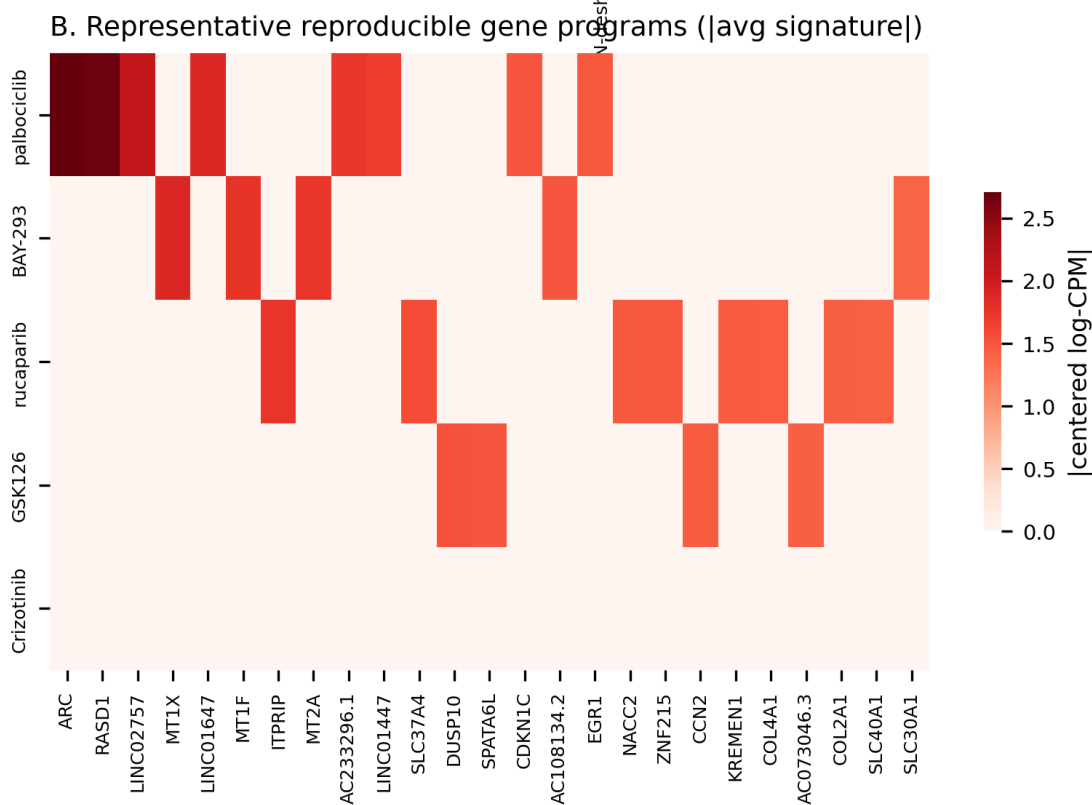
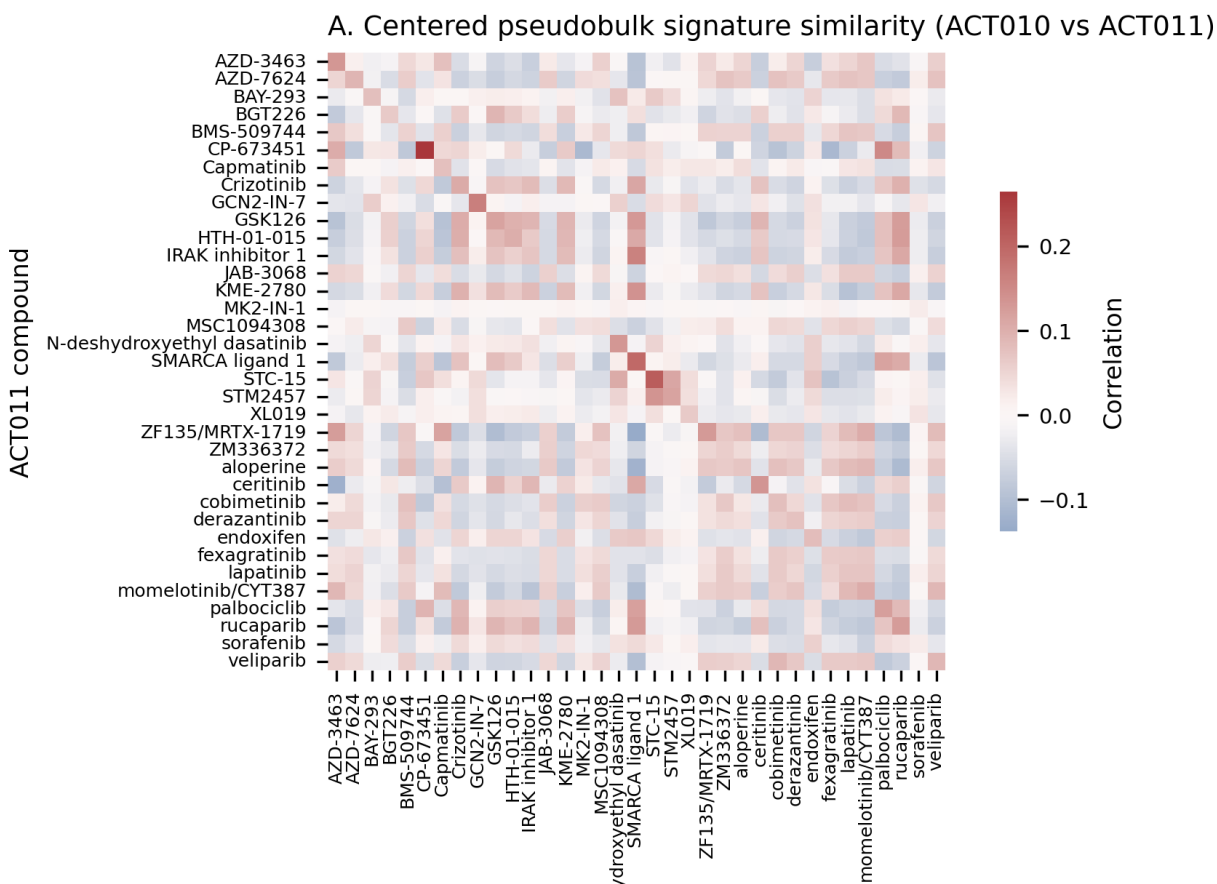


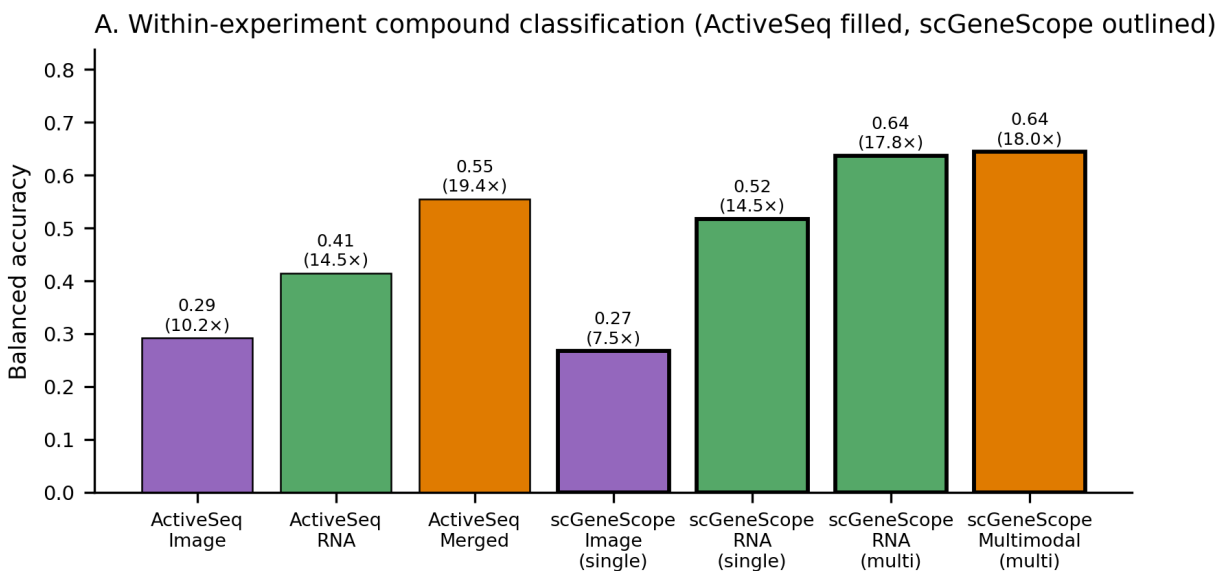
Figure 5. Centered pseudobulk gene signatures preserve compound-specific transcriptional programs across runs and recover the correct compound at 48.6% top-1 and 68.6% top-3 retrieval. The gene lists support RNA interpretability and triage rather than standalone mechanism calls.

scGeneScope provides external magnitude context

The final question is not whether ActiveSeq “beats” a public benchmark. The designs differ too much for that framing. scGeneScope profiles approximately 716,767 single-cell images and 627,704 single-cell RNA profiles across 28 treatments in U2-OS cells, with treatment-level matching across replicate populations [4]. This ActiveSeq pilot contains 11,511 control-panel matched wells across two HEK293 experiments, each read as a 5 to 10 cell pseudobulk, with a simpler imaging modality and a calibrated internal image latent.

The comparison is still useful as an order-of-magnitude check. ActiveSeq trails the strongest scGeneScope multimodal reference on raw within-experiment balanced accuracy, consistent with the scale, cell-line, and imaging-richness differences. Because the label spaces differ, fold over random-chance performance gives a rough normalization: pooled ActiveSeq RNA reaches 14.5 times chance, matching the scGeneScope single-profile RNA value; pooled ActiveSeq image reaches 10.2 times chance, compared with 7.5 times for the scGeneScope single-profile image value; pooled ActiveSeq merged features reach 19.4 times chance, compared with 18.0 times for the scGeneScope multimodal multiprofile reference.

The correct conclusion is restrained: a small same-well pilot reaches the same broad performance range after chance normalization, while measuring a different experimental unit. That supports continued scale-up. It does not establish superiority over scGeneScope or any imaging-first production system.



B. Context for interpreting the gap

Aspect	ActiveSeq (Z-Screen)	scGeneScope
Single-cell scale	~60-110k cells (5-10 × 11.5k wells)	~717k images, ~628k RNA
Label space	35 compounds	28 treatments
Pairing	Same matched micro-well	Treatment-matched replicate
Imaging	Brightfield + calibrated latent	5-ch Cell Painting + ViT
Pooled image	29.2% (10.2× chance)	26.7% single (7.5×)
Pooled RNA	41.4% (14.5× chance)	51.7% single (14.5×)
Pooled multimodal	55.5% (19.4×)	64.4% multi (18.0×)
Best ActiveSeq run	ACT010 merged 60.8% (21.3×)	n/a
Takeaway	Smaller pilot; same range after chance noise	Larger public benchmark on raw accuracy

Figure 6. Within-experiment compound classification for ActiveSeq versus the published scGeneScope reference, annotated with fold over random-chance performance, plus a context table summarizing the design differences that make the two benchmarks complementary rather than directly interchangeable.

Discussion

This pilot answers a foundational question, not the whole platform question. In 11,511 control-panel matched wells, both RNA and a calibrated image latent recover compound identity within and across runs. The two modalities share a compound map without collapsing into the same representation. When combined in a simple linear model, they improve held-out compound classification across essentially the whole panel. A same-compound shuffled-fusion control shows that this supervised gain is compound-level multimodal additivity rather than a measurable advantage of exact well-to-well pairing on the classification task.

The result is strongest at the compound level. Split-half reproducibility, cross-run centroid transfer, and supervised classification all depend on compound labels aggregated across many wells. They establish that ActiveSeq can produce a reproducible multimodal compound phenotype. They should not be described as proof that exact same-well pairing is the sole source of the gain. The within-compound analyses ask the stricter same-well question and are encouraging but modest: exact pair retrieval and distance-structure effects sit above random, but at small absolute magnitude. Larger replicate depth, stronger perturbational diversity, and prospective designs that enrich within-compound response heterogeneity are needed to determine when exact same-well pairing becomes operationally decisive.

The image result should be read with the same discipline. Calibration made compound-resolved image structure accessible, but the calibrated latent was learned on the same control-panel domain analyzed here. That is a useful internal representation for this pilot and a rational path for assay development. It is not yet a claim that the frozen image embedding or the calibration procedure will generalize unchanged to new libraries, cell types, or imaging runs.

For drug discovery, the most important downstream feature is that the RNA branch remains interpretable. The centered pseudobulk signatures do not provide definitive mechanisms of action, but they do preserve compound-specific gene programs well enough for triage: clustering hits, flagging broad stress programs, and nominating follow-up assays. This makes the multimodal gain more useful than a black-box classification result.

The competitive context is therefore directional. scGeneScope shows what is possible in a much larger treatment-matched public dataset with richer imaging and single-cell readouts. ActiveSeq reaches a similar broad range after chance normalization despite being smaller and pseudobulked, while measuring image and RNA from the same micro-well. The next experiments are clear: larger control panels, explicit unpaired-fusion controls, broader cell-line coverage, prospective chemistry beyond the calibration domain, and deeper analysis of whether same-well response-state structure becomes stronger with scale.

Methods

Dataset

The analysis uses the canonical Z-Screen ActiveSeq workspace under `data/ZScreen_Canonical_Dataset/`. Each ActiveSeq well is annotated with an experiment id (ACT010 or ACT011), a global well id, a compound assignment, an image embedding, and an RNA latent. Gene-level pseudobulk counts come from the matched H5AD object that backs the same-well RNA pipeline, repaired as documented in `DATA_REPAIR_REPORT.md`. The scGeneScope comparison uses published benchmark values transcribed from OpenReview Table 1 into `paper1a/tables/scgenescope_reference.csv`.

Per-well image embedding calibration

The original per-well image embedding was produced by a frozen pretrained vision model on raw imaging tiles. That representation captures generic visual structure but does not necessarily align compound identity with the principal directions of variance: experiment-of-origin, plate position, and microscopy artifacts can dominate compound signal in the raw geometry. Direct held-out compound classification on that embedding therefore tests the representation as much as the biology.

A 32-dimensional control-panel-calibrated latent was learned from the original embedding by a lightweight domain-adversarial network trained on the matched control panel. The objective rewards compound discrimination and penalizes experiment-of-origin discrimination, which pushes compound information onto the principal directions of variance without retraining a vision model from scratch. Calibration was learned on the same control-panel domain studied here, so the calibrated-latent results should be read as an internal representation refinement; broader-library and additional-cell-line generalization is the next test.

Matched-well preprocessing

Imaging detections were aggregated by `experiment_id` and `gwid` so each matched well contributed one image vector. Wells with an assigned control name were treated as the 35-compound panel. Wells without an assigned control name were the ZEL026 dummy-bead wells used for QC and were excluded from the compound-response analyses.

Reproducibility and transfer analyses

Within-run reproducibility was measured by repeated split-half pseudo-replication within each compound (cosine similarity between pseudo-replicate centroids after standardization within each matrix). Cross-run transfer used compound centroids learned in ACT010 and queried in ACT011.

Cross-modal geometry

For each run, pairwise distances among compound centroids were computed separately in image and RNA space. Agreement between distance matrices was summarized with Spearman correlation over upper-triangular entries, which does not assume the two scales are matched. Cross-run effect-size stability used Spearman correlation over per-compound effect magnitudes.

Supervised multimodal classification

Held-out classification used a 50:50 compound-stratified split, either pooled across runs or fit within each run. Linear discriminant analysis (eigen solver, automatic shrinkage) was trained on calibrated image features alone, RNA features alone, or their raw concatenation. LDA was used in preference to a deep classifier so that any reported gain is traceable to a linear function of the input features. Performance was summarized with overall accuracy, balanced accuracy, and per-compound recall. To test whether the supervised gain depended on exact image/RNA well pairing, a same-compound unpaired-fusion control shuffled RNA vectors among wells sharing the same compound and experiment before fitting the same merged classifier. The control used 100 shuffle iterations with the same train/test split.

Same-well representation learning

A lightweight CLIP-style dual encoder with a symmetric contrastive objective was trained on same-well pairs from the training split, evaluated by nearest-centroid compound retrieval and by exact held-out pair retrieval. Within-compound analyses compared image and RNA distance structure among wells assigned to the same compound and summarized distance-correlation, kNN-overlap, and same-compound pair-retrieval gains over random baselines. These analyses were treated as exploratory because the absolute effects were small.

Gene-level pseudobulk analysis

Raw counts were summed per experiment and control compound, converted to log-CPM, and centered by subtracting the experiment-wide mean signature. Cross-run retrieval used correlation similarity between centered signatures. Per-cell differential expression was not performed because each well averages 5 to 10 cells; pseudobulk plus centering is the unit at which the signal is robust at present depth.

External benchmark context

Within-run supervised metrics from `supervised_multimodal_metrics.csv` were compared to scGeneScope within-experiment values transcribed from OpenReview Table 1 into `scgenescope_reference.csv`. Because the label spaces differ (35 vs 28), the comparison reports both raw balanced accuracy and fold over random-chance performance.

Limitations

The pilot is intentionally narrow: one cell line (HEK293), 35 control compounds, and two matched experiments. The calibrated image latent was learned on the same control panel studied here; broader libraries and additional cell lines are needed to evaluate calibration generalization. The supervised fusion result shows additive value for combined image and RNA readouts, but the same-compound shuffled-fusion control indicates that exact well-to-well pairing does not add measurable classification value on this compound-label task. Exact same-well cross-modal retrieval and within-

compound structure remain modest in absolute terms. Gene-level signatures are summarized through pseudobulk aggregation rather than full differential-expression modeling with biological replicates. These caveats define the next analyses rather than negate the central conclusion: ActiveSeq supports reproducible compound-resolved multimodal profiling, with interpretable RNA and additive image-plus-RNA classification value.

Data availability

All data tables, derived latent representations, manuscript figures, and analysis inputs needed to reproduce this manuscript are organized in this repository under `paper1a/` and `data/ZScreen_Canonical_Dataset/`. A persistent-archive deposition with an assigned DOI will accompany the corresponding preprint posting.

Code availability

Analysis and figure-generation scripts are in `paper1a/scripts/`. Package-level dependencies are in the repository root `requirements.txt`. Tested with Python 3.14.3 on Windows.

References

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